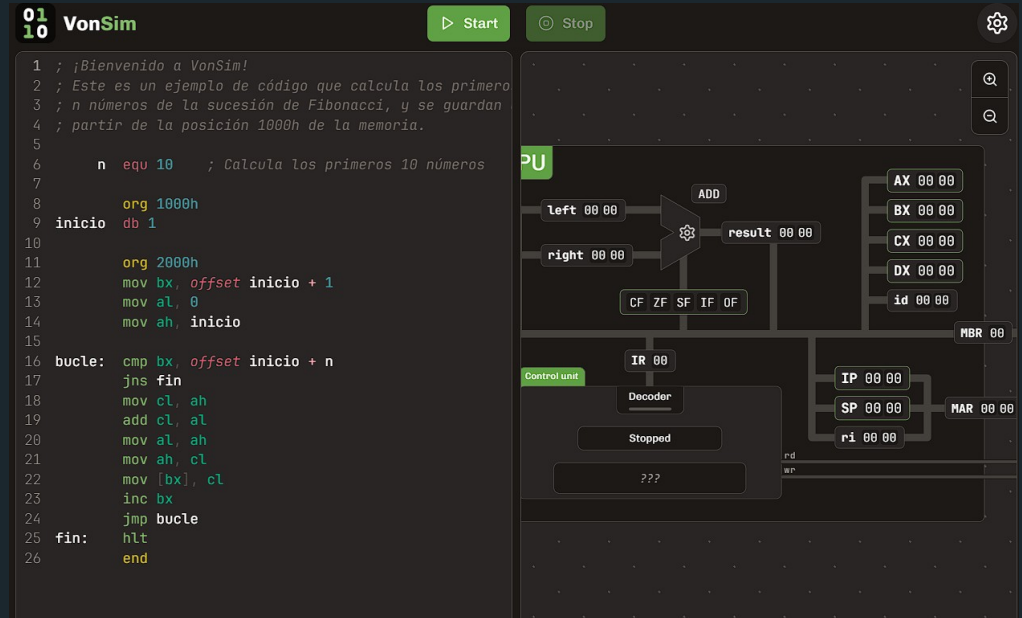




VonSim

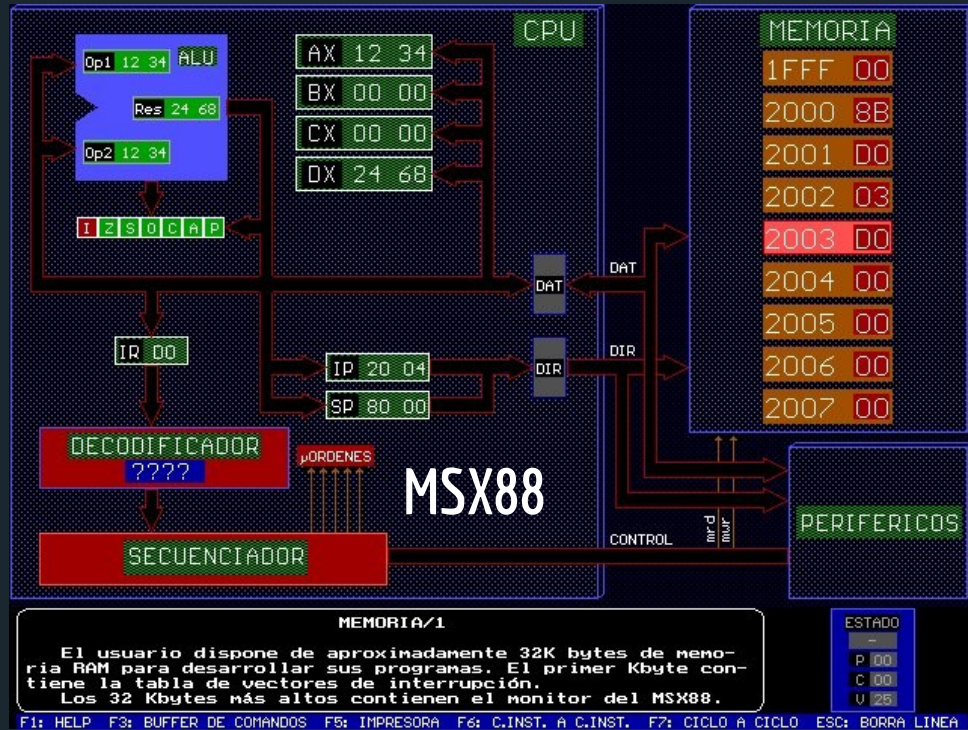
Educational Simulator for Assembly

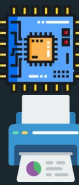


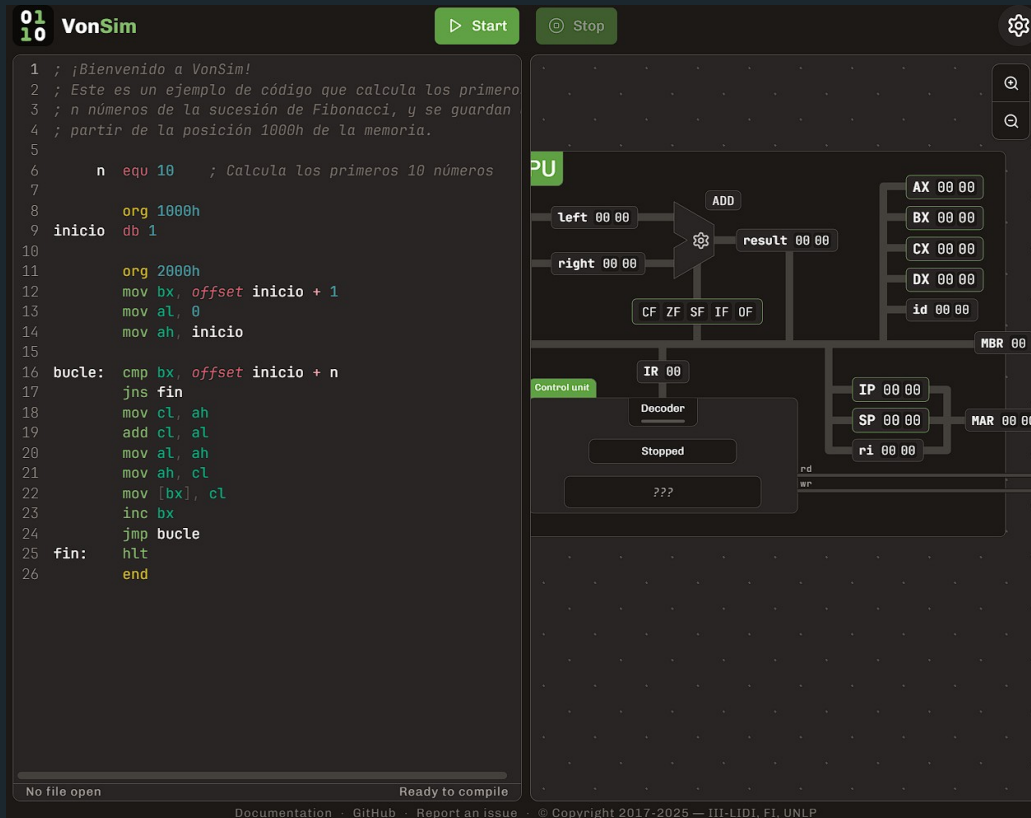


Context

- Informatics School
 - Universidad Nacional de La Plata
- Courses that teach Assembly
 - Computer Organization
 - Computer Architecture
- -1500 students per year
- Previously: **MSX88** simulator
 - VonSim 1: 2017
 - VonSim 2: 2023



- Web App 
- IDE 
 - Compile & Ex 
 - Debug 
- Simplified Architecture 
- Component Visualization 
 - I/O Devices 
- Integrated Examples 
- Mobile/Tablet mode 



The screenshot shows the VonSim web application interface. On the left, there is a code editor with assembly code. The code is as follows:

```

1 ; ¡Bienvenido a VonSim!
2 ; Este es un ejemplo de código que calcula los primeros
3 ; n números de la sucesión de Fibonacci, y se guardan
4 ; partir de la posición 1000h de la memoria.
5
6     n equ 10    ; Calcula los primeros 10 números
7
8     org 1000h
9 inicio db 1
10
11     org 2000h
12     mov bx, offset inicio + 1
13     mov al, 0
14     mov ah, inicio
15
16 bucle: cmp bx, offset inicio + n
17        jns fin
18     mov cl, ah
19     add cl, al
20     mov al, ah
21     mov ah, cl
22     mov [bx], cl
23     inc bx
24     jmp bucle
25 fin:  hlt
26      end
  
```

On the right, there is a visual representation of the VonSim architecture. It shows a CPU with various registers and components. The registers shown are AX, BX, CX, DX, and ID, each with a value of 00 00. The MBR (Memory Buffer Register) is also shown with a value of 00 00. The CPU is connected to a Memory Stack (MAR) and a Memory Buffer Register (MBR). The CPU is also connected to a Control unit, which includes a Decoder and a Stopped state. The CPU is also connected to a Memory Buffer Register (MBR) and a Memory Stack (MAR). The CPU is also connected to a Memory Buffer Register (MBR) and a Memory Stack (MAR).

At the bottom of the interface, there is a status bar that says "No file open" and "Ready to compile".

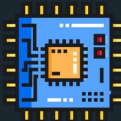


IDE

- Syntax Coloring
- Continuous compilation
- Detailed error messages with visual feedback

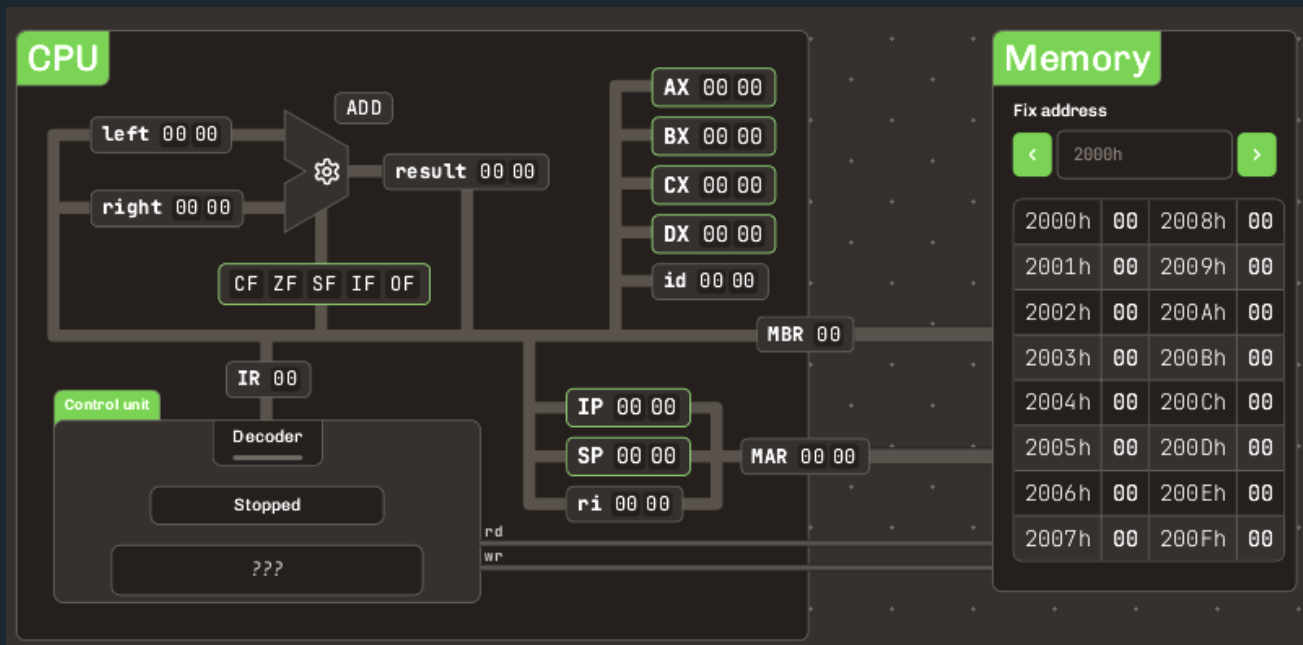
```
5
6 loop:    cmp bx, offset start + n
7          jns finish
8          mov cl, ah
9          add cl, al
0          mov al, ah
1          mov ah, cl
2          mov [bx], cl
3          inc bx
4          jmp loop
5 finish:  hlt
6          end
```

Label "LOOP" has not been defined.



CPU/Memory

- 8 half registers - 8 bits each: AL, AH, BL, BH, CL, CH, DL, DH
 - Can be addressed as 4 16-bit registers: AX, BX, CX, DX
 - Low/high parts
- Especial state registers
- ALU
- Memory: 16838 bytes
 - 16 bit addresses





Simplified 8088 Assembly

- **ORG <ADDRESS>**
 - Explicit starting address for code/data
- **END**
- **MOV dest, source**
 - Data transfers
- **OP dest[, source]**
 - OP = ADD | SUB | OR | AND | XOR | NEG | NOR
- **hlt**
 - Stop execution

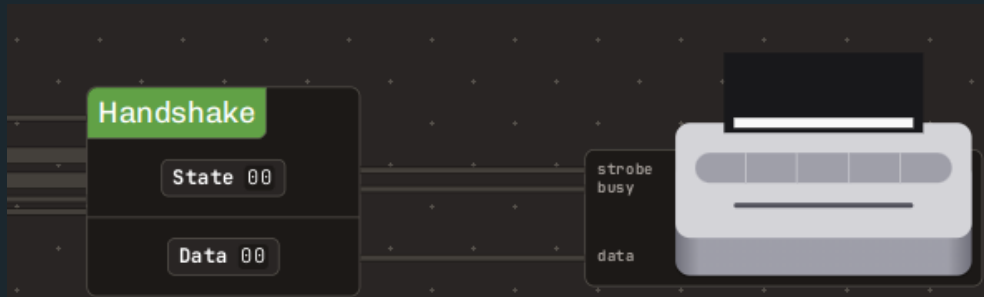
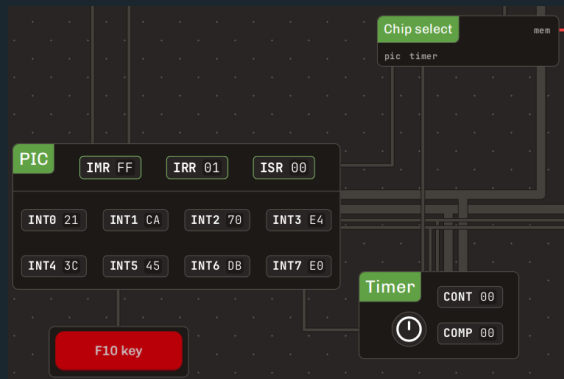
Compute $C = A + B$

```
1 | ; DATA
2  org 1000h
3  A  db 5
4  B  db 3
5  C  db ? ; no init
6
7  ;2000h: default address
8  ; to start executing code
9  org 2000h
10 mov al, A
11 add al, B
12 mov C, al ; C = A + B
13 hlt
14 end
```

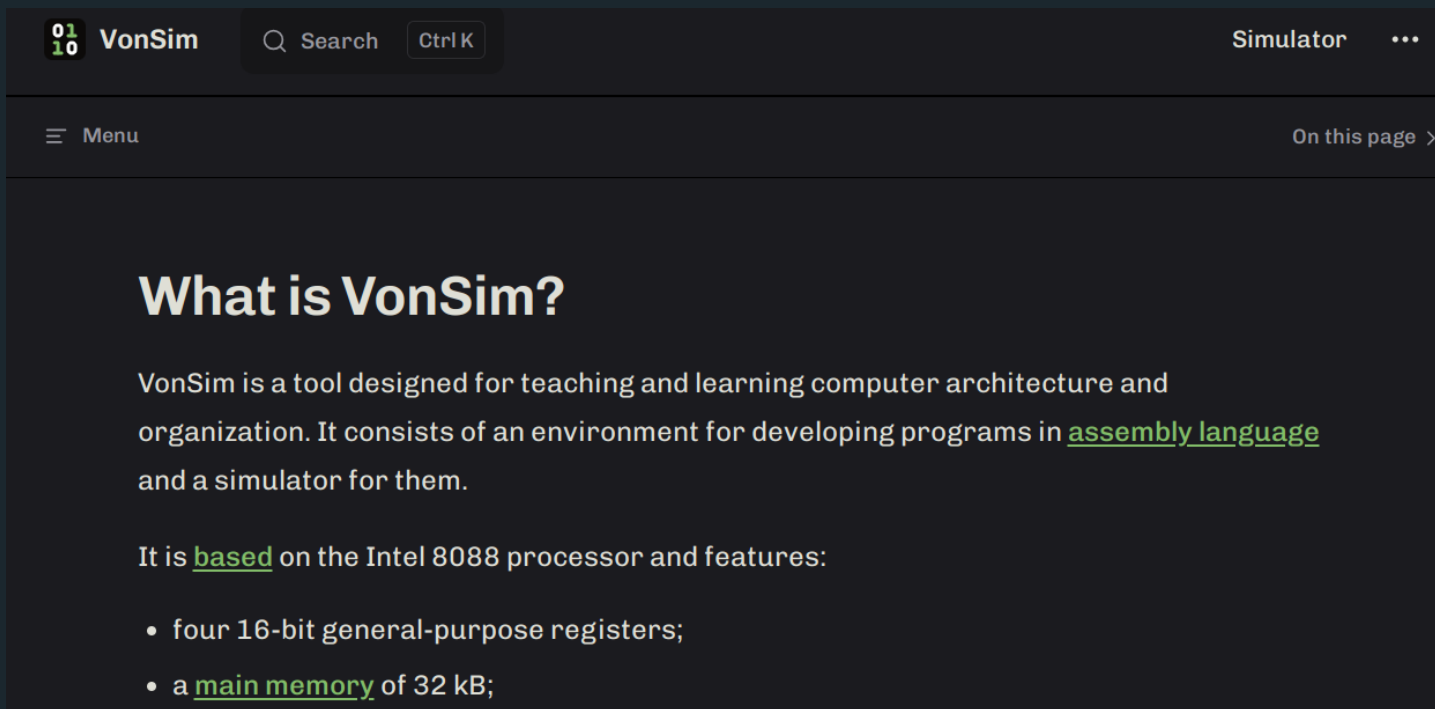
Device I/O



- Keyboard & Screen
- PIC
 - → Timer
 - → F10 Button
 - → HANDSHAKE
- PIO
 - → LEDs & Switches
 - → Printer
- HANDSHAKE
 - → Printer



VonSim 2 Docs



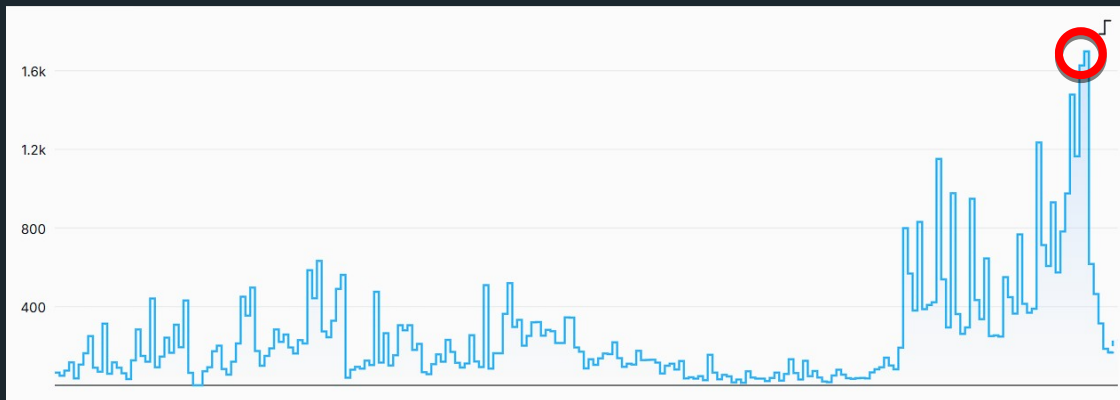
<https://vonsim.github.io/en/>



VonSim 2 Usage statistics

- Via OneDollarStats (2025/03 - 2025/10)

Visits per day



Devices

Devices		Browser	OS	Device
Device type		Visits		
Desktop		27.1k		
Mobile		6.29k		
Tablet		174		
Unknown		35		

VonSim 2 Team



Juan Seery

<https://github.com/JuanM04>



Facundo Quiroga

<https://github.com/facundoq>



<https://github.com/vonsim/vonsim>